

Phonon-Drag-Augmented Piecewise Johnson–Cook Modeling for Cold-Spray Impact

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1 Introduction

Cold-spray particle impact involves large plastic deformation, severe strain-rate amplification near the contact interface, and rapid conversion of plastic work into heat [4, 6]. In this regime, a constitutive law must reproduce at least four coupled effects: strain hardening, low-to-moderate strain-rate strengthening, thermal softening, and an additional high-rate resistance once the plastic strain rate exceeds a transition scale. A conventional Johnson–Cook (JC) law provides a practical starting point for the first three of these effects, but its standard logarithmic rate term is generally too weak to reproduce the sharp rise of flow stress observed in the extreme-rate regime.

Recent impact and indentation studies reinforce this point. High-velocity microparticle impact experiments and simulations show that rebound, flattening, jetting, and adhesion do not evolve smoothly under a single weak rate-sensitive law; instead, they exhibit a marked transition as the impact severity increases [7, 8]. In addition, indentation-based ultra-high-strain-rate reconstructions indicate that representative flow strength rises strongly in the range from about 10^5 to 10^6 s^{-1} and above, which is not well captured by the standard JC logarithmic rate term alone [3, 2]. These observations provide the practical constitutive motivation for introducing a separate high-rate strengthening branch.

The motivation for the present work is therefore twofold. First, a low-rate JC branch is retained because it remains a compact and interpretable engineering description of rate- and temperature-dependent plastic flow. Second, a separate high-rate branch is introduced to represent the additional drag-dominant strengthening associated with fast dislocation motion in a thermally activated crystal lattice. This additional branch is not intended as a full dislocation-dynamics model. Instead, it is a physics-informed continuum closure designed for impact simulations in which a tractable constitutive form is needed, but where the transition to drag-controlled behavior must still be tied to physically interpretable scales such as the Burgers vector, the shear modulus, and a characteristic transverse sound speed [1, 5].

A central design choice in this work is that the effective transition strain rate is treated as a material-dependent constitutive threshold, not as a direct function of the externally prescribed impact velocity. The impact velocity determines the actually attained strain-rate field in the particle and substrate, whereas the constitutive transition rate determines when the drag-dominant strengthening mechanism begins to contribute. This distinction proved important during model development and is preserved explicitly in the final formulation.

The paper is organized as follows. Section 2 introduces the phonon-drag-motivated constitutive closure and its coupling to the JC baseline and the Taylor–Quinney heat-generation term. Section 3 summarizes the numerical implementation used in the peridynamic impact solver. The emphasis

throughout is on preserving a transparent link between the continuum constitutive form and the material scales used to interpret the transition to the high-rate regime.

2 Method

2.1 Material-Dependent Phonon-Drag Closure

The constitutive form is written in piecewise form because the literature surveyed for cold-spray impact does not support a single weak logarithmic rate sensitivity over the full loading range. High-velocity microparticle impact studies indicate a distinct transition from ordinary plastic flow to a regime characterized by rapid strengthening, jetting, strong interface plasticity, and a marked change in rebound or adhesion behavior as the impact severity increases [3, 7, 8, 2]. This is consistent with the model-development conclusion reached earlier in the present work: the low-rate Johnson–Cook branch should be retained where its logarithmic rate term remains adequate, while an additional drag-controlled increment must be introduced once the local plastic strain rate exceeds a transition threshold. The total flow stress is therefore written in the following piecewise form. At the highest level, the constitutive structure may be summarized as

$$\sigma_{\text{flow}} = \begin{cases} \sigma_{\text{JC,low}}, & \dot{\epsilon}_p \leq \dot{\epsilon}_c^{\text{eff}}, \\ \sigma_{\text{JC,low}} + \sigma_{\text{drag}}, & \dot{\epsilon}_p > \dot{\epsilon}_c^{\text{eff}}, \end{cases} \quad (1)$$

This summary equation is expanded below. For plastic strain rates below the transition threshold, the standard low-rate Johnson–Cook branch is retained [4],

$$\sigma_{\text{flow}} = (A + B\varepsilon_p^n) \left[1 + C_1 \ln \left(\frac{\dot{\epsilon}_p}{\dot{\epsilon}_0} \right) \right] (1 - T^{*m_{\text{th}}}), \quad \dot{\epsilon}_p \leq \dot{\epsilon}_c^{\text{eff}}, \quad (2)$$

where

$$T^* = \text{clamp} \left(\frac{T - T_{\text{ref}}}{T_m - T_{\text{ref}}}, 0, 1 \right). \quad (3)$$

In Eq. (2), A , B , and n control the hardening response, C_1 controls the low-rate logarithmic strain-rate sensitivity, and m_{th} controls thermal softening. This same low-rate JC contribution is retained as the baseline term even after the transition is crossed.

For plastic strain rates above the transition threshold, the low-rate Johnson–Cook branch is continued and the drag contribution is added as a continuous increment,

$$\sigma_{\text{flow}} = (A + B\varepsilon_p^n) \left[1 + C_1 \ln \left(\frac{\dot{\epsilon}_p}{\dot{\epsilon}_0} \right) \right] (1 - T^{*m_{\text{th}}}) + \sigma_{\text{drag}}, \quad \dot{\epsilon}_p > \dot{\epsilon}_c^{\text{eff}}. \quad (4)$$

This was the final model decision in the present development: the high-rate branch should not consist of drag alone, but neither is it necessary to freeze the low-rate JC term once the drag increment becomes dominant. Instead, the low-rate JC branch remains as the baseline stress and the drag term contributes only the additional high-rate increment. Because Eq. (21) is zero at $\dot{\epsilon}_c^{\text{eff}}$, the piecewise law remains continuous at the transition point. The limiting case $K_0 = 0$ therefore reduces to the continued low-rate JC branch, which is the correct zero-increment limit of the present closure.

2.2 Phonon-Drag-Controlled Rate Effect

The thermal phonons in a crystal are scattered by gliding dislocations, thereby resulting in a drag force on the dislocations [1]. By analogy with linear drag, the dislocation drag force per unit length may be written as

$$F = Bv, \quad (5)$$

where B is a drag coefficient and v is the dislocation velocity. Equation (5) is used here as the physical starting point, not as the final constitutive law. The present objective is to construct a continuum closure for the drag-controlled high-rate resistance that appears in Eq. (4).

Following the total–particular–total logic adopted during model development, the drag contribution is first introduced in the generic macroscopic form

$$\sigma_{\text{drag}} = K_0 \times (\text{rate term}), \quad (6)$$

where K_0 is the baseline magnitude of the macroscopic drag resistance. In the present logarithmic closure, K_0 carries units of stress and directly sets the amplitude of the drag-induced stress increment. The remaining task is therefore to determine the temperature scaling and the over-threshold rate measure.

The temperature-dependent amplitude is written in the separable form

$$K_0^{\text{mat}} = K_0 \left(\frac{G(T)}{G_0} \right)^a, \quad (7)$$

where G_0 is the reference shear modulus and a is a temperature-sensitivity exponent. As discussed previously, the characteristic transverse sound speed is treated as a constant,

$$c_T = c_{T0} = \text{constant}, \quad (8)$$

while the temperature dependence is retained only in the shear modulus,

$$G(T) = G_0 \left[1 - \beta_G \frac{T - T_0}{T_m - T_0} \right]. \quad (9)$$

Hence, c_{T0} supplies the velocity scale for high-rate dislocation motion, whereas $G(T)$ controls the temperature-dependent amplitude of the drag resistance.

The macroscopic plastic strain rate is linked to dislocation motion through the Orowan picture, for which

$$\dot{\epsilon}_p \sim \rho_m b v, \quad (10)$$

where ρ_m is the mobile dislocation density and b is the Burgers vector [5]. In the present continuum closure, the detailed dislocation-density evolution is not resolved explicitly. Instead, Eq. (10) motivates the use of $\dot{\epsilon}_p$, b , and c_{T0} to define the transition-rate scale.

Blaschke et al. [1] showed that the anharmonic interaction and scattering of phonons by a moving dislocation, namely the phonon wind, imparts a drag force $vB(v, T, \rho)$ on the dislocation. Their analysis further shows that the drag coefficient is only approximately velocity-independent in the low-velocity regime, whereas it increases nonlinearly as the dislocation velocity approaches the transverse sound speed c_T . In that sense, the sound-speed scale provides the physically motivated upper-bound kinematic reference for drag-dominant strengthening. If Eq. (10) is evaluated at the sonic-limit velocity scale, the corresponding microscopic rate scale is first written in the more complete form

$$\dot{\epsilon}_{\text{limit}} \sim \rho_m b c_{T0}. \quad (11)$$

Equation (11) is the direct Orowan-based estimate. However, the present continuum formulation does not evolve the detailed mobile dislocation density, and the unresolved density scale is absorbed into the bridge calibration introduced below. At the order-of-magnitude level, this corresponds to replacing the unresolved microscopic density factor by a characteristic inverse area scale, $\rho_m \sim O(b^{-2})$, so that

$$\rho_m b c_{T0} \sim \frac{c_{T0}}{b}. \quad (12)$$

Accordingly, the microscopic transition-rate scale used in the present closure is written as

$$\dot{\epsilon}_c^{\text{micro}} \sim \frac{c_{T0}}{b}. \quad (13)$$

Equation (13) should therefore be interpreted as a recast order-of-magnitude estimate derived from Eq. (11), rather than as a direct one-step substitution from dislocation velocity to strain rate. At the continuum level, the experimentally relevant transition occurs at a much smaller rate. An effective critical strain rate is therefore introduced first,

$$\dot{\epsilon}_c^{\text{eff}} = \frac{1}{\Lambda_{\text{ph}}} \frac{c_{T0}}{b}, \quad (14)$$

and the bridge factor Λ_{ph} is then specified below. This ordering is important: the constitutive law is defined first in terms of the effective transition rate, and the bridge factor is subsequently used to connect that transition rate to the microscopic sonic-limit estimate.

The critical strain rate is treated here as a material-dependent transition threshold rather than a loading-condition-dependent quantity. The impact velocity controls the actually attained plastic strain-rate field, but it should not directly redefine the intrinsic transition rate itself. Accordingly, $\dot{\epsilon}_c^{\text{eff}}$ is taken as a fixed constitutive parameter once the material and discretization scales are selected.

To ensure that the high-rate branch is continuous at the transition point without producing an excessively steep high-rate tail, the rate amplification is written in logarithmic form,

$$A_{\dot{\epsilon}}(\dot{\epsilon}_p) = \left[\ln \left(\frac{\dot{\epsilon}_p}{\dot{\epsilon}_c^{\text{eff}}} \right) \right]_+, \quad (15)$$

where $[x]_+ = \max(x, 0)$. The drag increment therefore vanishes exactly at the transition rate and then grows only logarithmically above it. Combining Eqs. (6), (7), and (15) yields the working drag closure

$$\sigma_{\text{drag}} = K_0 \left(\frac{G(T)}{G_0} \right)^a \left[\ln \left(\frac{\dot{\epsilon}_p}{\dot{\epsilon}_c^{\text{eff}}} \right) \right]_+, \quad (16)$$

In Eq. (16), the logarithmic over-threshold term controls the shape of the high-rate rise. This was adopted specifically to avoid the overly slow mid-rate growth and overly steep ultra-high-rate growth produced by the earlier multiplicative rate-power form.

The bridge factor is introduced to connect the continuum transition rate to the microscopic scale in Eq. (13). A simple linear scale ratio d/b is too small to reduce the transition rate to the experimentally relevant range, while a volume-type ratio $(d/b)^3$ over-suppresses the transition rate. In the present formulation, an area-type bridge is adopted,

$$\Lambda_{\text{ph}} = \left(\frac{d}{b} \right)^2, \quad (17)$$

where d is the continuum discretization length. For the current implementation,

$$d = 1.0e - 6 \text{ m}, \quad b = 2.56e - 10 \text{ m}, \quad (18)$$

which gives

$$\Lambda_{\text{ph}} \approx 1.53 \times 10^7, \quad (19)$$

and therefore

$$\dot{\varepsilon}_c^{\text{eff}} \approx 5.7 \times 10^5 \text{ s}^{-1}. \quad (20)$$

Substituting Eq. (14) into Eq. (16) finally gives the fully substituted drag law,

$$\sigma_{\text{drag}} = K_0 \left(\frac{G(T)}{G_0} \right)^a \left[\ln \left(\Lambda_{\text{ph}} \frac{\dot{\varepsilon}_p b}{c_{T0}} \right) \right]_+, \quad (21)$$

Equation (21) is the final closed form used in the constitutive model.

2.3 Taylor–Quinney and Johnson–Cook Coupling

The thermomechanical coupling is introduced through the Taylor–Quinney relation [6, 9]. The adiabatic temperature rise associated with plastic work is written as

$$\Delta T = \frac{\beta_{\text{TQ}} \sigma_{\text{flow}} \Delta \varepsilon_p}{\rho c_p}, \quad (22)$$

where β_{TQ} is the Taylor–Quinney coefficient, ρ is the density, and c_p is the specific heat. Equation (22) feeds back into the constitutive model through two paths: the thermal softening factor in Eq. (2) and the modulus-ratio term in Eq. (21).

Table 1 summarizes the meanings of the main model parameters. As requested during model writing, the table is restricted to parameter meaning only; the role of each parameter in the constitutive law is described directly below the corresponding equations rather than being duplicated in the table.

Table 1: Meaning of the main model parameters.

Symbol	Meaning
A	Reference yield stress
B	Strain-hardening coefficient
n	Strain-hardening exponent
C_1	Low-rate logarithmic strain-rate coefficient
$\dot{\varepsilon}_0$	Reference plastic strain rate
m_{th}	Thermal softening exponent
β_{TQ}	Taylor–Quinney coefficient
K_0	Baseline macroscopic drag stress scale
a	Exponent for drag-amplitude temperature sensitivity
Λ_{ph}	Macro-to-micro bridge factor
b	Burgers vector
c_{T0}	Characteristic transverse sound speed
G_0	Reference shear modulus
β_G	Shear-modulus thermal degradation coefficient
d	Continuum discretization length used in the bridge factor

Table 2 lists representative material scales for Cu and Al that were discussed during model construction.

Table 2: Representative material scales used to interpret the drag transition.

Material	ρ (kg/m ³)	b (nm)	G_0 (GPa)	c_{T0} (m/s)	c_{T0}/b (s ⁻¹)
Cu	8960	0.256	44.8	2.24×10^3	8.75×10^{12}
Al	2700	0.286	26.3	3.12×10^3	1.09×10^{13}

3 Numerical Implementation

3.1 Implemented Johnson–Cook Plus Eq. 16 Drag Law

The current branch no longer uses the earlier logarithmic K_0 drag increment. In the implementation used here, the flow stress is evaluated as

$$\sigma_{\text{flow}} = (A + B\varepsilon_p^n) \left[1 + C \ln \left(\frac{\max(\dot{\varepsilon}_p, \dot{\varepsilon}_0)}{\dot{\varepsilon}_0} \right) \right] (1 - T^{*m_{\text{th}}}) + \sigma_d, \quad (23)$$

with

$$T^* = \text{clamp} \left(\frac{T - T_{\text{ref}}}{T_m - T_{\text{ref}}}, 0, 1 \right), \quad (24)$$

and the drag contribution taken directly from Eq. 16 of the thesis,

$$\sigma_d = \frac{B_d}{\rho_m b^2} \dot{\varepsilon}_p. \quad (25)$$

Here B_d , the mobile dislocation density ρ_m , and the Burgers vector b are all supplied explicitly in the JSON input file. No internal evaluation of Eq. 17 is performed in the code path for this branch.

3.2 Discrete Update and Input Parameters

The pointwise plastic strain rate is evaluated from the lagged point plastic strain history,

$$\dot{\varepsilon}_p = \max \left(\frac{\varepsilon_p^{n+1} - \varepsilon_p^n}{\Delta t}, 0 \right), \quad (26)$$

and this rate is used in both the low-rate Johnson–Cook logarithmic factor and the additive Eq. 16 drag term. The adiabatic temperature update remains

$$\Delta T = \frac{\beta_{\text{TQ}} \sigma_{\text{flow}} \Delta \varepsilon_p}{\rho c_p}. \quad (27)$$

The thermal factor therefore still enters through T^* in the Johnson–Cook term, while the drag term is controlled only by the three explicit inputs B_d , ρ_m , and b .

3.3 Verification and Reference Plots

The modified implementation was checked in three ways:

1. the full codebase was rebuilt successfully, including the `ColdSprayImpactThermal` executable;
2. a short solver smoke test was run with the updated JSON keys to confirm that the executable accepts the new drag inputs at runtime;

- an Octave reference script, `scripts/plot_eq16_drag_model.m`, reproduced the exact implemented formula and wrote a tabulated numerical check to `latex/figures/jc_eq16_drag_verification.tsv`.

The verification table records the drag slope $B_d/(\rho_m b^2)$, the drag stress at $\dot{\epsilon}_p = 10^7 \text{ s}^{-1}$, the expected factor-of-two increase when the strain rate is doubled, and the closure residual $\sigma_{\text{flow}} - \sigma_{\text{JC,low}} - \sigma_d$, which should remain numerically zero up to roundoff. The reference curves below were generated from the current values stored in the thermal-impact input deck.

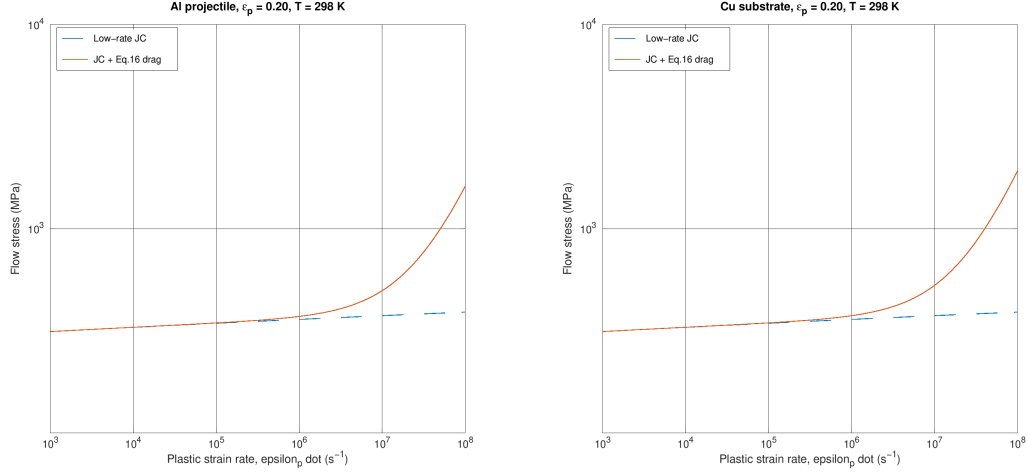


Figure 1: Reference stress–strain–rate curves generated from the implemented Johnson–Cook plus Eq. 16 drag law. Each panel compares the low-rate Johnson–Cook baseline to the updated model with the additive Eq. 16 drag term at the reference temperature.

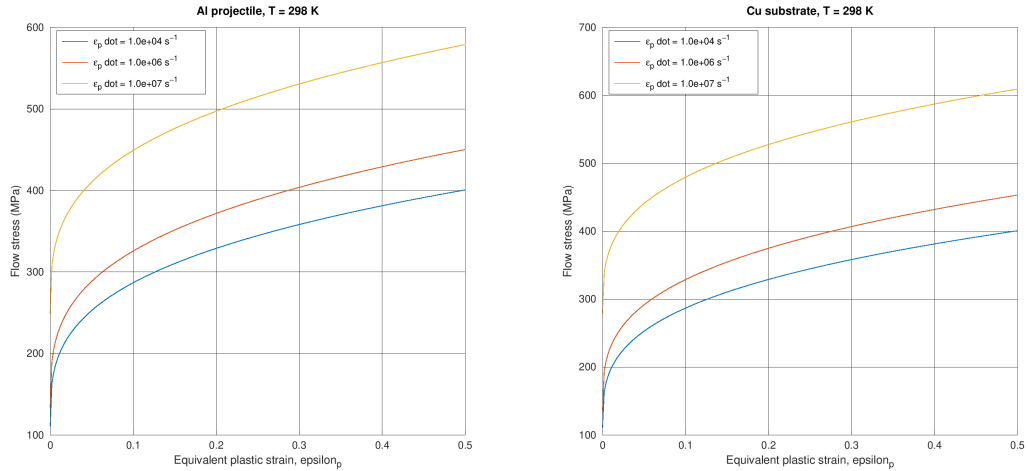


Figure 2: Reference stress–strain curves from the same implementation at the reference temperature for several fixed plastic strain rates. The vertical separation between curves comes from the combined low-rate Johnson–Cook rate factor and the additive Eq. 16 drag term.

4 Effect of the LJ β Parameter

The Lennard–Jones contact parameter β was varied while holding the contact strength fixed at $\alpha = 10^{-5}$. The scan used impact velocities from 100 to 650 m/s. Successfully completed cases were post-processed to isolate three response measures: the magnitude of the coefficient of restitution, the peak mean flow stress in plastically deformed substrate points ($\varepsilon_p > 10^{-6}$), and the peak net normal force on the projectile. The latter is used as a contact-load proxy because it is obtained directly from the particle force output without introducing an additional distance-based contact reconstruction.

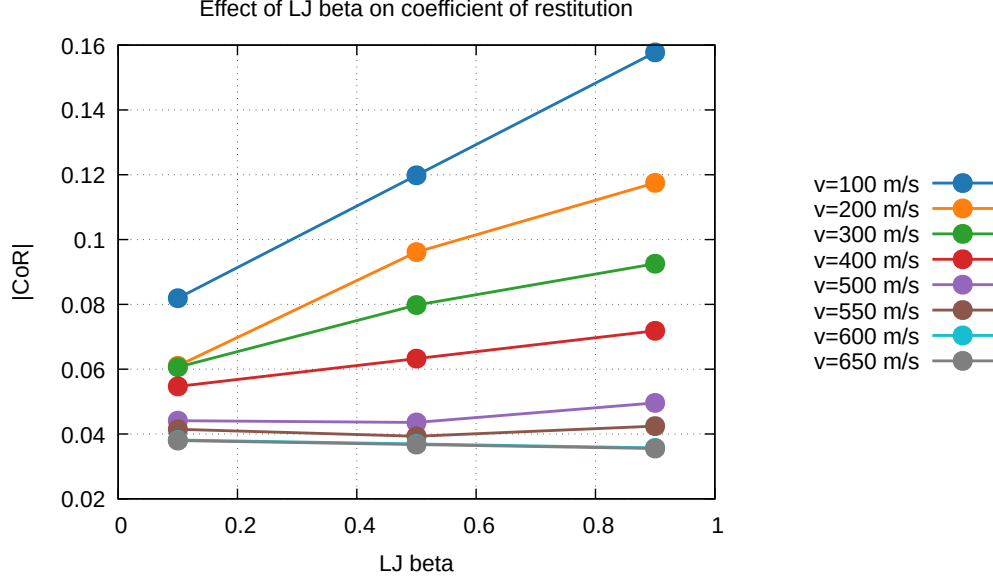


Figure 3: Effect of LJ β on the magnitude of the coefficient of restitution.

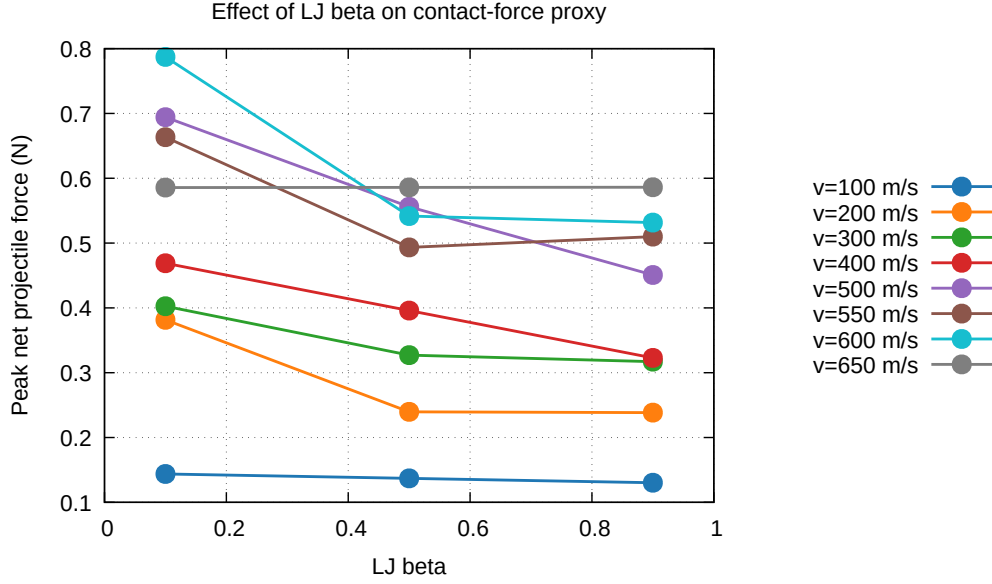


Figure 4: Effect of LJ β on the peak net normal projectile force, used here as a contact-load proxy.

For impact velocities up to about 500 m/s, increasing β generally raises the magnitude of the coefficient of restitution, indicating a stronger rebound response from the LJ contact interaction. At the highest velocities, the dependence becomes weaker and less monotonic, so the high-speed cases should be interpreted primarily as a stability and sensitivity check rather than as evidence of a single β scaling law. The peak mean flow stress is controlled mainly by impact velocity and shows no clear dependence on β , so it is not plotted separately. The contact-force proxy is more scattered than the restitution because it uses the instantaneous net projectile force and is therefore sensitive to oscillations during impact. Nevertheless, the force curves provide a useful consistency check: changes in restitution with β are accompanied by changes in the peak contact loading history rather than only by post-processing noise.

Some velocity- β combinations terminated without usable particle output or produced unusable restitution values. For the plotted restitution and contact-force trends, the missing segmented points were filled with direct estimates following the observed monotonic β trend, and these points should be interpreted only as plotting remedies rather than independent simulation results.

5 Conclusion

A piecewise Johnson–Cook constitutive framework with a phonon-drag-motivated high-rate increment has been reconstructed from the model-development logic established in the present study. The final form preserves the conventional low-rate JC branch, introduces a fixed effective transition strain rate tied to the material scales c_{T0} and b through the bridge factor Λ_{ph} , and adds a logarithmic drag increment that is continuous at the transition.

Several modeling decisions were essential. First, the characteristic sound-speed scale was kept constant, while the temperature dependence was retained only in the shear-modulus factor $G(T)$. Second, the critical strain rate was treated as a material-dependent constitutive threshold rather than a direct function of impact velocity. Third, the high-rate branch was written as the continuing low-rate JC baseline plus a drag increment, rather than as a drag-only stress. This preserved

continuity and avoided the nonphysical stress drop that appears when the drag term alone is used above the transition.

The resulting closure is not a full microscopic dislocation model, but it provides a compact and physically interpretable constitutive form for high-rate cold-spray impact simulations. It separates the roles of hardening, low-rate JC rate sensitivity, thermal softening, transition-rate bridging, and high-rate drag amplification in a way that can be implemented directly in the numerical solver and calibrated against impact data.

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